

# Principles and Interactive Tools for Evaluating and Improving the Behavior of Natural Language Processing models

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## ABSTRACT

While the accuracy of Natural Language Processing (NLP) models has been going up, users have more expectations than captured by just accuracy. Despite practitioners' attempt to inspect model blind spots or lacking capabilities, the *status-quo* processes can be ad-hoc and biased. My thesis focuses on helping practitioners organize and explore the inputs and outputs of their models, such that they can gain more systematic insights into their models' behaviors. I identified two building blocks that are essential for informative analysis: (1) to scale up the analysis by grouping similar instances, and (2) to isolate important components by generating counterfactuals. To support multiple analysis stages (training data assessment, error analysis, model testing), I designed various interactive tools that instantiate these two building blocks. In the process, I characterized the design space of grouping and counterfactual generation, seeking to balance the machine powers and practitioners' domain expertise. My future work proposes to explore how the grouping and counterfactual techniques can benefit non-experts in the *data collection* process.

## CCS CONCEPTS

• **Human-centered computing** → Empirical studies in HCI; Interactive systems and tools; • **Computing methodologies** → Machine learning.

## KEYWORDS

Natural Language Processing; Error Analysis; Training Data Assessment

### ACM Reference Format:

Tongshuang Wu. 2021. Principles and Interactive Tools for Evaluating and Improving the Behavior of Natural Language Processing models. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3443423>

## 1 INTRODUCTION

**Context and motivation.** While the accuracy of Natural Language Processing (NLP) models has been going up, users have more expectations than captured by just accuracy. For example,

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*CHI '21 Extended Abstracts*, May 8–13, 2021, Yokohama, Japan

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ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3443423>

chatbot assistants should not make inappropriate or unfair replies to certain kinds of inquiries, and translators should not strive for perfect performances in English by sacrificing their support on low-resource languages. To guard the end-user experience, experts and stakeholders (those train and deploy models) are responsible for making informed decisions, deploying trustworthy models while iterating on those that have clear weak spots. However, state-of-the-art models are inherently complex, which makes it hard for model developers to understand if a model is behaving as expected or where the model insufficiency lies, let alone making targeted improvements.

Various studies have attempted to help practitioners interact with their models in an iterative fashion [2]: the current model explains itself by presenting sample predictions and evidence, and practitioners in turn give model feedback, starting the cycle again. Proponents argue that through such rapid exchange, practitioners can make focused observations, and can drive models towards desired behaviors with quick feedback. However, through human-subject experiments [3, 35], I observed that local contexts may not provide enough information, and may lead practitioners towards false trust or suboptimal interventions.

**Research objective & method.** My research aims to help NLP practitioners understand and improve their models more globally and systematically. In particular, I focus on *enabling practitioners to comprehensively validate the inputs and outputs of their models*, as such validations encourage more unbiased observations, and dissuade practitioners from making uninformed decisions. For example, by assessing the training data quality, practitioners can spot biases the model is learning from, and provide root fixes via data augmentation; by enumerating multiple error causes in model failure analysis, practitioners can optimize their future improvement by prioritizing more severe model weaknesses. More concrete research questions include:

- Q1 What components are essential for systematically analyzing model-related data?
- Q2 How can we build tools with these components for different analysis stages, such that practitioners can easily apply their domain expertise based on the inherent properties of NLP tasks and datasets?
- Q3 How can such components be beneficial more broadly — in processes beyond model analysis (e.g., model design and training), and for non-expert users (e.g., hired labors in data annotations)?

**Current Contributions.** Inspired by literature surveys and close interactions with domain experts, I have *identified building blocks that are essential for informative analysis* [34] (Q1): First, to make precise and scalable observations, we need a *quantifiable*

*grouping method* that finds a *collection* of similar instances. Second, to highlight the core elements (e.g., the most impactful features that determine the label), we need a *counterfactual perturber* that isolates the effect of certain patterns by making targeted minimal rewrites.

I instantiated these two building blocks for different analysis stages (Q2) and, in doing so, *explored the balance and the integration of machine powers and human intelligence*. When facilitating error analysis [34], I introduced a domain-specific language to help experts maximally express their domain knowledge; Whereas in the case of data quality assessment, to prioritize the inherent properties of the datasets, I further extended the design of grouping methods, such that the system can automatically mine representative patterns, and suggest potential data groups that were worth of further exploration [36].

**Next steps and expected contributions.** In the line of enhancing and applying the two building blocks (Q2), I am currently exploring the design space of perturbation methods. I have surveyed the desired properties for different downstream tasks of counterfactual generation, and plan to design methods that can increase the diversity and creativity of perturbations. Further, in addition to *understanding existing data*, I plan to explore how grouping and counterfactual perturbation benefit the process of *collecting new data*, by shaping annotators' behaviors in crowdsourcing tasks (Q3). I expect the techniques to encourage end-users to generate more creative datasets, and to label the examples more effectively and consistently. Ultimately, I expect my work to help support multiple model development stages that involve inputs to and outputs of machine learning models.

**Background, Dissertation status and long term goals** I am a fifth-year Ph.D. student in the Paul G. Allen School of Computer Science and Engineering at the University of Washington, working with Professor Jeffrey Heer and Professor Daniel S. Weld. I conduct research as part of the Interactive Data Lab and Human-AI Interaction Lab. My work focuses on helping practitioners effectively and systematically interact with their models, by organizing and exploring the data input to and output from machine learning models. I am currently preparing for my thesis proposal, and expect to finish in 2022. Upon graduation, I plan to continue the research work in academia or industrial research labs.

## 2 BACKGROUND AND RELATED WORK

### 2.1 Local explanation and intervention on models

Various work has encouraged local understandings and interventions on models. For example, local explanations has been used to express why a model made a specific prediction. These are often given in terms of input features (e.g., LIME [26], SHAP [22], anchors [27]) or groupings of model weights [14, 19]. While they are useful for helping end users develop intuitions of model behavior, they usually do not reflect global patterns, and may even mislead users to overlook related contexts. In a user study investigating the human-model team performance (where the human classify documents based on models' recommendations) [3], I found that humans performed worse when the model highlighted a sentence that justifies its (incorrect) recommendation as the local explanation. It

seems they were too convinced by the mere "rationale" sentence to notice the opposing evidence in the rest of the document.

Similarly, while rapid and focused model updates supported by Interactive Machine Learning (IML) [2, 5, 20, 31] is designed to drive models toward intended behaviors through intuitive trial-and-error, I found that incremental iterations can be dangerous for the model update. When I asked participants to improve a sentiment analysis model by adding or removing individual word features for 50 iterations as they saw fit [35], I observed that such local feature selection (a technique supported and advocated for in IML research) failed to improve models on average, and sometimes hampered generalization on the test set.

These experiments sound cautionary notes, and paved the way for my subsequent research. First, local contexts may not provide enough information, and may lead practitioners towards false trust or suboptimal interventions. Conversely, I designed systems that present a comprehensive picture of models to enhance unbiased understanding. Second, step-wise tuning may be harmful to model performance, and be a poor utilization of human effort. To achieve effective improvement, I plan to encourage more stable and reusable contributions via data correction or augmentation.

### 2.2 Status-quo Analysis process

Prior studies generally agree on two common forms of model analyses: First, *data grouping*, where aggregate metrics are computed for particular slices of interest (e.g., accuracy over question types in machine comprehension) [13, 21]. We expect examples in the same group to share similar patterns, such that the model performance on the group can reveal the general trend of model behaviors. Second, *counterfactual analysis*, where one slightly modifies the input data to eliminate potential spurious feature-label correlations [16, 28]. We expect counterfactual perturbations to fulfill the gap between limited linguistic variations in the training data and the diversity in real-world languages [18, 32].

In practice, however, with the input to most NLP tasks being unstructured text, grouping and counterfactual analysis are coarse and limited. Groups created manually or via slicing scripts [25] are usually not representative of the dataset distribution. Filters on text properties can scale (e.g., word frequency [9], sentence length or answer type [17, 33]), but prior work has noted that such features usually could either overlook the context of the whole sentence [23, 30], or get too abstractive to be precise and interpretable [6]. In other words, typical grouping struggles with the trade-off between being reproducible/scalable, and being in-depth and meaningful. I address such issues by adding structure to the text data based on linguistic structures and queries on textual patterns. Similarly, counterfactual attacks to models either rely on manual rewrites [11, 18] — and therefore are diverse yet unscalable — or take limited, mostly rule-based forms of word substitutions, keyword deletions, or paraphrasing [1, 10, 24]. Moreover, they are mainly for the purpose of detecting over-stability or over-sensitivity. In contrast, I am interested in generating counterfactual examples that are diverse yet systematic, for the purpose of understanding *why* models behave in certain ways.

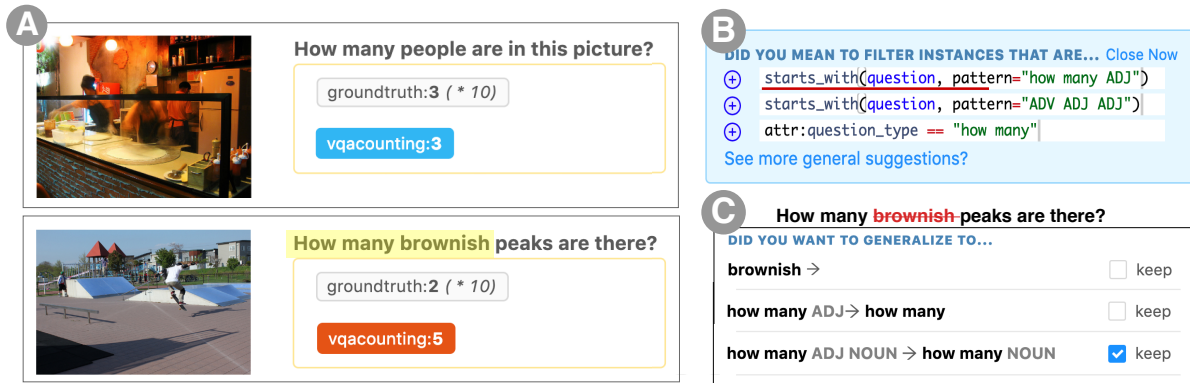


Figure 1: Errudite example: A visual question answering model predicts “how many people” correctly, but is incorrect on “how many brownish” (A). Experts suspect the token following “how many” can make a difference. They generalize the observation into groups (B) capturing patterns of “how many NOUN” and “how many ADJ”, and test the hypothesis by removing the differing adjectives (C).

### 3 RESEARCH PROGRESS TO DATE

#### 3.1 Building blocks for informative analysis: Grouping and Counterfactual analysis

In practice, NLP developers usually analyze when, how, and why models fail by manually annotating a subset of error samples with their potential error causes. However, the approach is laborious and subjective, which can lead to high variance and low reproducibility. Moreover, by focusing on error cases independent of situations where the model is correct, the analysis can yield biased results. Finally, since multiple linguistic features usually co-exist, the root cause of errors can easily be overlooked.

I summarized that a successful error analysis process should be precise, reproducible, scalable, and testable. In particular, rather than error subsets, we should *slice the entire dataset into large groups*, so to scale the analysis to cover all the relevant successes and failures. On the other hand, rather than individual examples, we should *focus on paired examples with isolated, contrasting patterns* to expose the root causes of errors. Accordingly, I presented a tool called *Errudite* [34] to enable the application of these principles. Errudite concretizes unambiguous error hypotheses into grouping queries based on an expressive domain-specific language (DSL) (Fig. 1B). It also generates counterfactual examples through rewrite rules, which supports testing the error hypotheses (Fig. 1C). In user studies, I observed that Errudite enabled users to perform high quality and reproducible error analyses with less effort. Errudite also revealed substantial ambiguities in prior published error analysis practices. When given identical descriptions of an error type and asked to reproduce it, our users produced groups that vary in size from 13.8% to 45.2% of all errors.

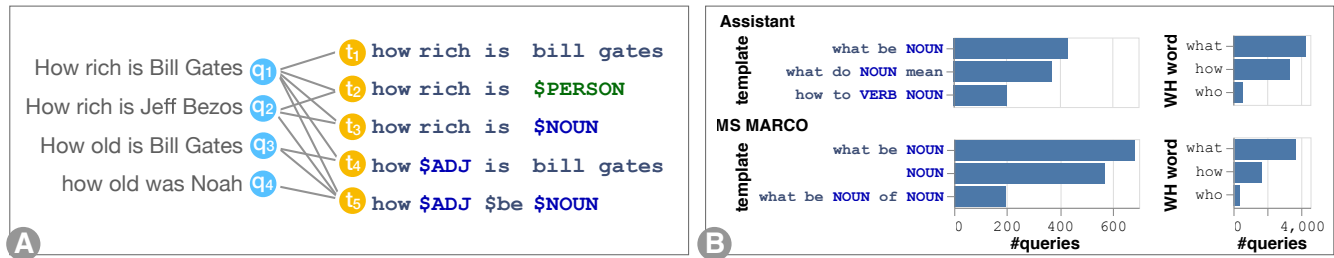
Having identified the importance of grouping and counterfactual analysis in Errudite, my collaborator and I further presented CheckList [29], a framework and a tool that *facilitates comprehensive model evaluation*. CheckList went one step further to make suggestions on *where* to group or perturb. It enumerates linguistic capabilities and expected outcomes that users may overlook otherwise, so to guide them design test case groups and perturbations

with high test coverage. Practitioners with CheckList found almost three times as many bugs as those without it.

#### 3.2 Structural Templates: Extending the grouping to prioritize dataset properties

One fundamental functionality in Errudite is to systematically gather and modify instances within groups. In fact, data grouping is also essential for dataset exploration, in which practitioners make sure that their sampled data meet their modeling needs (e.g., having required labels and metadata) and do not have any data quality issues (e.g., conflicting labels on similar instances). However, existing methods for finding data groups are insufficient. Errudite’s manual compilation of filters relies heavily on users’ domain expertise, which can be prone to human errors. On the other hand, fully automated methods like text clustering can produce groups not semantically meaningful to humans.

To help inspire actionable and unbiased analysis, my collaborators at Apple Inc. and I developed a tool called *Tempura* [36]. Tempura supports analyzing queries from search engines and intelligent assistants, using interpretable, context-preserving, representative, and flexible groups recognized by a new grouping strategy — structural templates. As in Figure 2, structural templates allow analysts to create query groups from different structural similarity aspects, at different granularities. Based on these query groups, Tempura facilitates the analyses by showing overviews first and providing details on demand. To offer a starting point for the analysis, it automatically selects a template subset that best represents the query distribution. To help analysts navigate the template space, it also suggests related templates likely to yield further explorations. Processing both academia and industrial query datasets with Tempura, I observed that structural templates could reveal dataset distributional differences. User studies also showed that Tempura can support both data exploration and error analysis, helping practitioners examine the distribution of the data, find labeling errors, and discover model error patterns and outliers in a query dataset. Experts appreciated the efficient and intuitive categorization. As



**Figure 2: (A) Structural templates are generated by replacing tokens with abstract ones based on linguistic features. For instance,  $t_2$  abstracts the query  $q_1$  by replacing the token “Bill Gates” with its named entity type, **PERSON**. Templates offer text groupings from different aspects. While  $t_2$  uses **PERSON** to find rich celebrities, one can instead get  $t_4$  by replacing “rich” in  $q_1$  with **ADJ** to explore other queries about “Bill Gates” (e.g., “how old” in  $q_3$ ). (B) The three most representative templates from Tempura, and most frequent wh-words for the sampled questions in two datasets. The datasets have similar query types, but their templates differ after “what be NOUN”. This indicates that the templates can capture more dataset characteristics.**

one user described, “I learned 7 new things about the dataset and the model in the last half an hour! Without the tool, I would spend all those time writing Python scripts without knowing if it will lead to anything significant.”

## 4 FUTURE WORK

### 4.1 Ongoing work: Extending the perturbation for diverse but controlled counterfactual reasoning

While Tempura extends the characteristics of grouping, in an ongoing work, I further explore the design space of perturbation methods. Similar to grouping, prior work on counterfactual generation typically follows one of two extremes. On the one hand, templates or perturbation rules (used by Errudite and Checklist) allow targeted inspections (Figure 1C), but can only cover limited linguistic patterns. On the other hand, those that thrive in diversity are either too uncontrolled (e.g., text generation [15]) or hard to scale (e.g., manual rewrites [11, 18]).

I seek to achieve a balance between control and generation diversity. To this end, I have fine-tuned language models for perturbation generation. To enable targeted inspection, I allow practitioners to control *how* to change a sentence (with perturbation types defined based on common linguistic capabilities), and *where* to change (based on linguistic features like part-of-speech tagging or dependency trees.) For example, to test the impact of negation, instead of enumerating rules like “did → didn’t”, “VERB → would never VERB”, one can say “add negation modifiers to aux.”

Moving forward, I plan to explore whether our method is helpful in various downstream tasks. First, the perturbations can potentially serve as extensive counterfactual explanations, i.e., explaining models’ reaction to a set of changes, rather than one change [8, 27, 28]. A group of highly related changes may reveal model insufficiencies that are hard to spot otherwise (e.g., sentiment analysis model only recognizing certain kinds of negations like “did not”, but misclassifying others like “I would never.”) We plan to verify the insightfulness of these explanations through typical evaluation methods, e.g., surprise, simulatability [12]. Second, I hope to collect more effective contrast sets [11] that can reveal the vulnerability of the model. I

plan to annotate the generated perturbations with class labels in crowdsourcing tasks (e.g., positive or negative in sentiment analysis). Then, to evaluate, I plan to verify whether we can find more bugs in the state-of-the-art models (i.e., if the accuracy further drops on these datasets). Further, the data collected can be used for data augmentation [18]. I plan to explore whether models trained with such perturbations can become more capable of handling certain capabilities (e.g., pass more tests in Checklist related to negation).

### 4.2 Next step: Extending groupings and perturbations to data collection

My prior research has focused on how grouping and perturbations can be used to assess data related to NLP models. As a next step, I am interested in how the methods can improve the data collection, such that instead of applying structures “after the fact”, we can provide valuable groupings and counterfactual reasoning at the time of capture. These metadata can supply finer-grained model training (e.g., used as constraints in model structures and help build models that are “right for the right reasons”) and analysis with less downstream effort. More importantly, we expect these methods to reshape crowdworkers’ labeling behavior, which may help improve the quality of the labeled data.

For example, in training data collection, data grouping at the labeling time can potentially improve the labeling consistency. By associating each labeled instance with e.g., certain grouping filters (as in Errudite) or templates (as in Tempura), we can easily spot other instances that share similar semantics, and flag the ones that are labeled inconsistently. If we further translate such filters and templates into Snorkel labeling functions [7, 25], we can increase the labeling efficiency, by suggesting automatic labels for patterns that always result in the same class labels. On the other hand, in counterfactual dataset creation, we expect our perturbation generator to help spark labelers’ creativity by asking them to create more counterfactual examples that are different from those covered in automatic generation.

With the annotators usually being hired laborers, a challenge is to explore interaction strategies that can flexibly translate intuitive non-expert interactions into rich forms of metadata. For example, I

Figure 3: A preliminary interface in the perturbation study, to collect labels for the automatically generated counterfactuals.

used a simple form of programming by demonstration in Errudite to ease the process of formulating group filters and rewrite rules, which can be further enriched: If we label examples together with their perturbed pairs, it is possible to automatically infer groups based on the impact of perturbations (e.g., examples where adding or removing negations will result in a different class label.) The system can further try to confirm its guesses with the annotators in an interactive fashion (as a back and forth conversation), such that they can “query” users’ intents more precisely. [4]

## ACKNOWLEDGMENTS

I would like to thank my collaborators, the Interactive Data Lab and the Human-AI Interaction Lab at the University of Washington, and the many others who have supported me in my research thus far. Special thanks to my advisors, Jeffrey Heer and Dan Weld. This work was supported by the Moore Foundation Data-Driven Discovery Investigator program, with additional support from ONR grant N00014-18-1-2193, the WRF/Cable Professorship, the Allen Institute for Artificial Intelligence and Google.

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